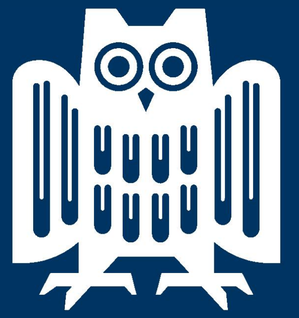
Spatiotemporal Modeling of Citizen Tree Watering in Berlin

Weather Responses and Neighborhood Variation

Universität des Saarlandes, Interdisciplinary Institute for Societal Computing
Thesis Proposal for M.Sc. Data Science & AI

Joseph Carpenter Sheils | Supervisor: Angela John
Master Seminar · October 2026



Overview

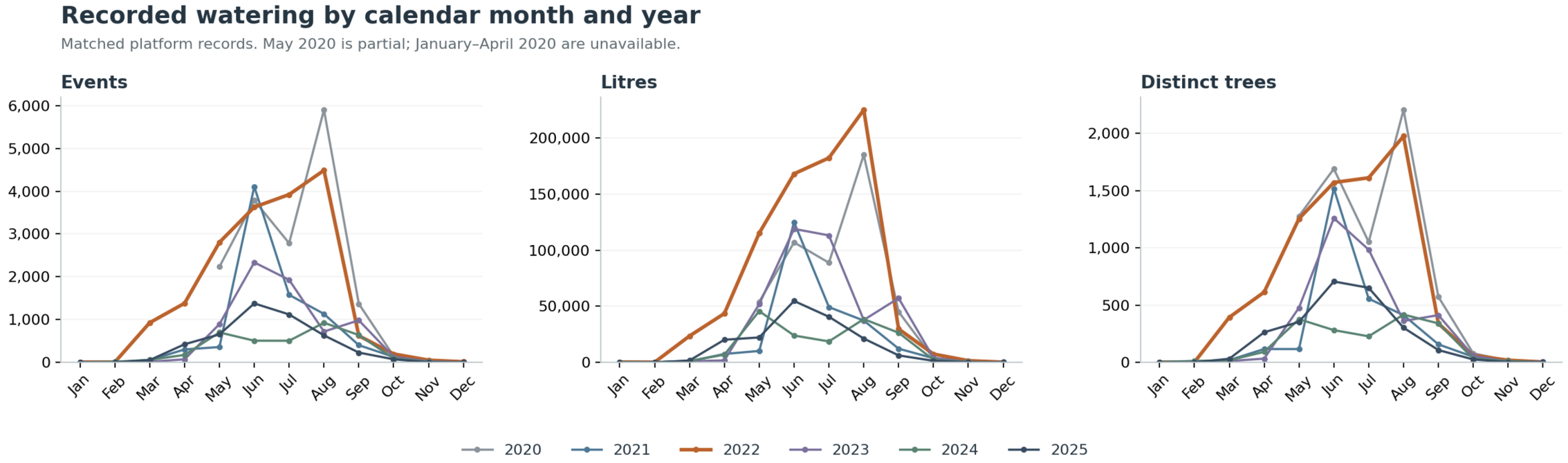
1. Introduction & Background
2. Research Gap and Questions
3. Planned Methodologies
4. Conclusions + Bibliography

Introduction & Background

Abstract + Motivation

Tree cooling depends on water; citizen care varies across Berlin.

About 59,000 platform records connect weather, tree characteristics and neighbourhood context. The core question is how logged care responds to heat and dryness.



Background

Conceptual Risk Framework (IPCC)

Hazard	Daily maximum air temperature, dry-spell length and drought anomaly
Surface exposure	Morning land surface temperature and baseline thermal load
Sensitivity / capacity	Social status and the share of residents aged 65+
Observed response	Change in logged watering as heat and dryness increase
Ecosystem service	Canopy and low-vegetation cooling, and its loss on dry days

Civic heat response is a behavioural indicator relevant to adaptive capacity; it is not a direct measure of capacity.

Related Works

Vegetation cooling and soil moisture

Schwaab et al. (2021); Gobatti et al. (2025): cooling varies with vegetation, climate and water availability.

Unequal cooling loss

Yan et al. (2025): drought-related vegetation and cooling losses differ across US communities.

Behavioural adaptation and participation

Stechemesser & Wenz (2023); Kirklies et al. (2024): heat responses and access to digital participation can be unequal.

Irrigation decision support

QTrees; Rebholz et al. (2026): estimate tree water needs and motivate a comparison that also accounts for logged volunteer care.

Research Gap and Questions

Research Gaps

1 Cooling loss within a European city

Estimate who experiences drought-related cooling loss in Berlin, separating tree canopy from low vegetation.

2 Collective civic heat response

Use daily records of care for shared infrastructure to study how neighbourhood responses differ—and whether participation explains the difference.

3 Allocation around volunteer effort

Compare municipal irrigation priorities based on need alone with priorities that account for expected logged citizen care.

The scope of the allocation gap will be checked through a structured literature search in Month 1.

Research Questions

RQ1 · Cooling loss | Supporting study

How much does vegetation cooling weaken on dry days, and is total loss greater in areas with lower social status and higher baseline LST?

RQ2 · Civic heat response | Core study

Does logged watering respond to daily heat and dryness, and do social differences remain after accounting for baseline participation?

RQ3 · Allocation | Exploratory study

Can logged care of young street trees be predicted under spatial and temporal hold-out, and how do the two allocation strategies rank areas?

Bridge: Does stronger civic heat response occur where more vegetation cooling is lost?

Hypotheses

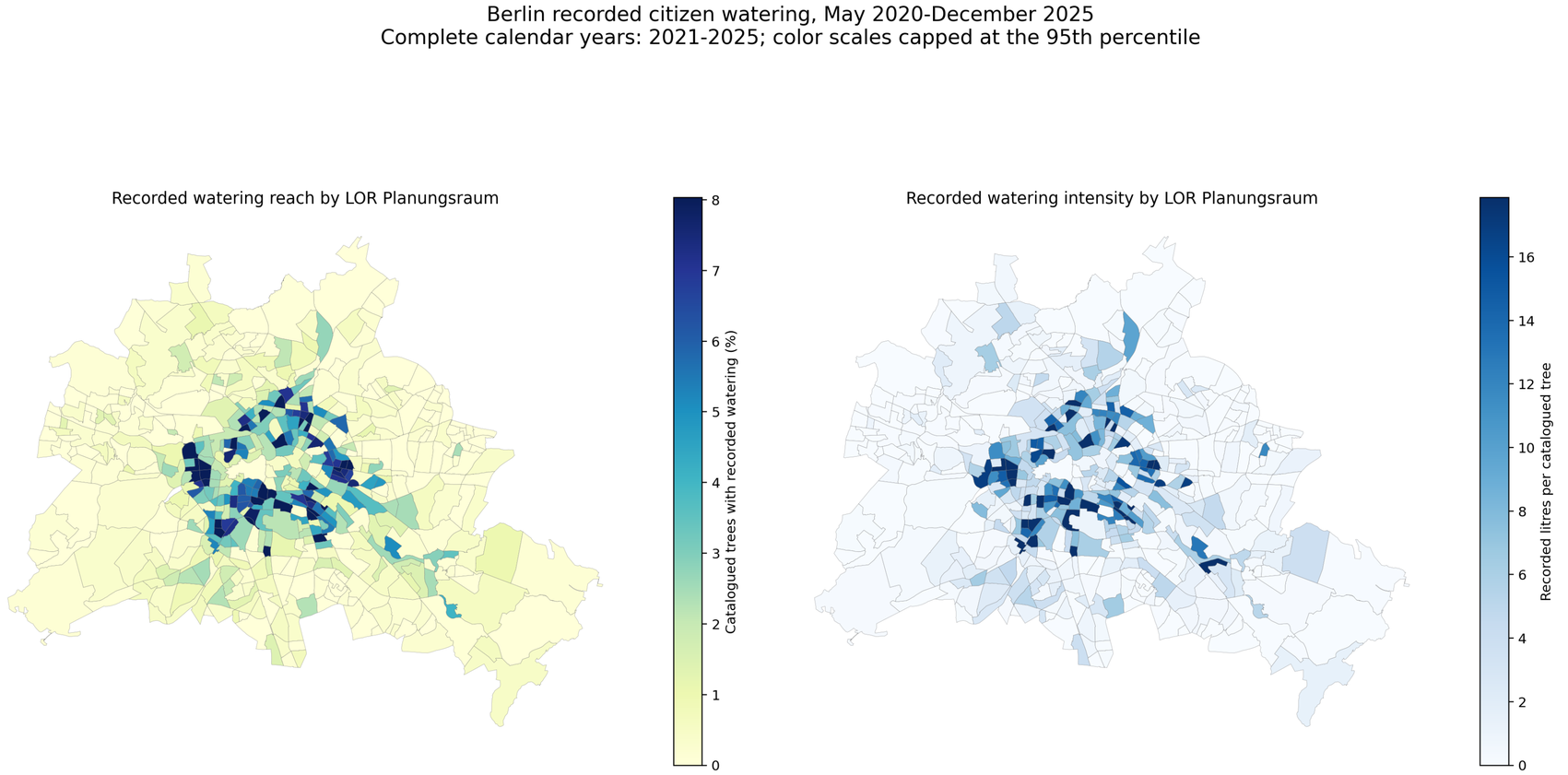
H1 · Cooling loss	Greater total loss on dry days in areas with lower social status and higher baseline LST.
H2 · Unequal response	Weaker heat- and dryness-response slopes in areas with lower social status.
H2-alt · Participation	The social gradient is explained by baseline participation and participation proxies.
H3 · Static benchmark	Weak agreement with MSS status: rank correlation < 0.5 and $\geq 20\%$ of areas in off-diagonal terciles.
H4 · Bridge	Civic heat-response slopes are uncorrelated or negatively correlated with total cooling loss.

Planned pre-registration before modelling. H2 must persist after baseline participation, age structure and turnout are considered.

Planned Methodologies

Study area & Units of Analysis

Berlin: 12 districts · 542 LOR Planungsräume · records from May 2020 to December 2025



Part I

Satellite cells → planning areas

Part II

Planning-area days; 143 Bezirksregionen as fallback

Part III

Tree-seasons → area priorities

Part 1: Cooling Loss and its Incidence

Outcome and vegetation

Morning Landsat LST, 2017–2024; canopy and low-vegetation fractions from vegetation height, with dated Sentinel-2 NDVI.

One hierarchical model

Vegetation × temporal drought × social status; date effects, area slopes and built-form controls carry uncertainty into the equity comparison.

Quantity reported

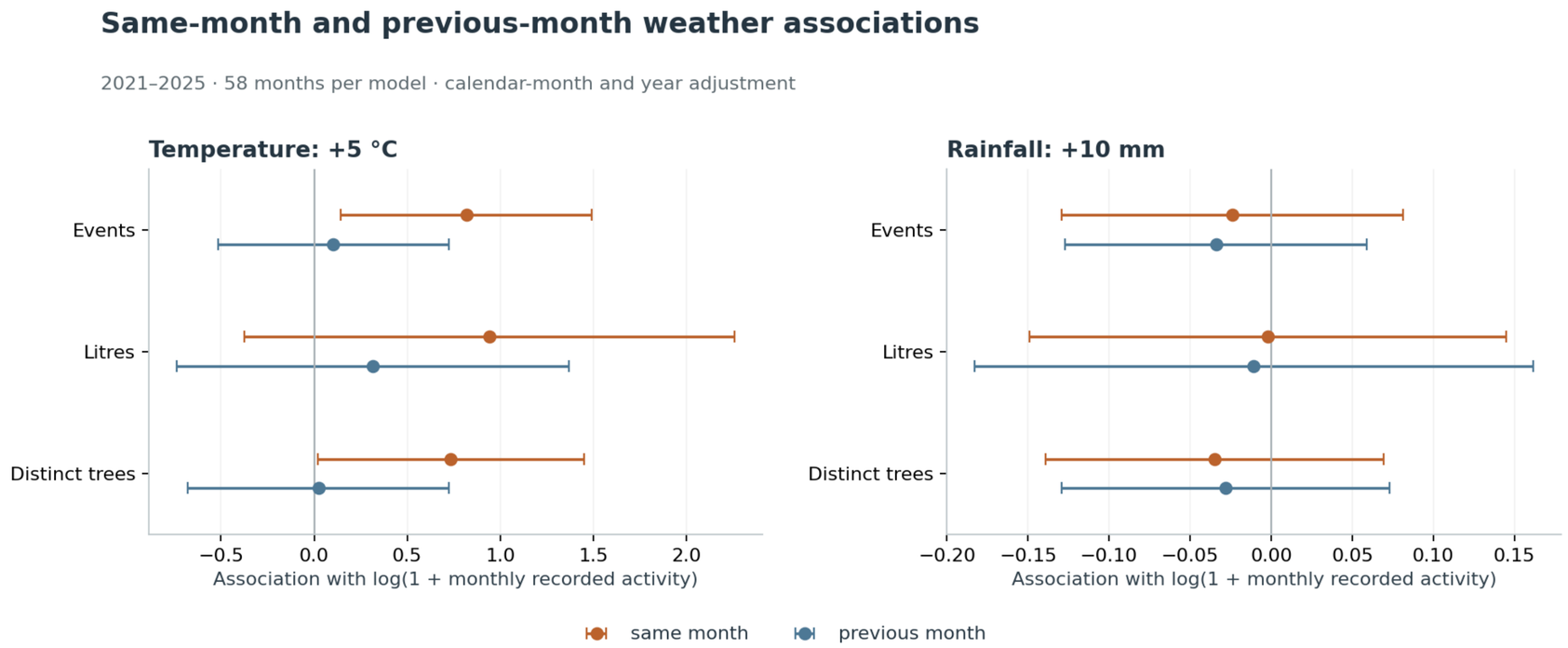
Per-unit loss: change in the dry-versus-wet vegetation slope. Total cooling loss: per-unit loss × local vegetation fraction.

Checks: sealed-cell placebo · spatially blocked validation · residual spatial dependence · ECOSTRESS sensitivity where available

Part 2: Civic Heat Response (core study)

Participation: logistic model of whether an area ever records care.

Daily response: hierarchical negative binomial with young-tree offset, area intercepts and heat/dryness slopes. Adjust for weather, calendar, year and baseline participation.



Why a daily model?

The monthly heat association motivates a daily test of timing and neighbourhood variation.

Figure: exploratory monthly results

Part 3: Allocation around volunteer effort (exploratory)

Predict logged care for street trees aged 5–19

Regularised logistic regression and gradient boosting; label: any recorded watering in a tree-season.

Validate before ranking

Block by planning area; train on 2020–2023 and test on 2024–2025. Report PR-AUC, AUROC, calibration and Brier score.

Expected unmet water = max(0, seasonal dose – P(logged care) × mean litres if logged)

Priority = need × expected unmet water

Compare need-only and complement-based rankings by rank correlation and top-50 overlap, with sensitivity to weights.

Predicts logged care. The choice between allocation strategies remains a policy question.

Work Plan and Feasibility

Month 1	Audit and reconcile counts; literature check; data requests; pre-register hypotheses.
Months 1–2	Build the area-day behavioural panel and the satellite–vegetation panel.
Months 2–4	Fit the core civic-response model; test participation explanations and stability.
Months 3–4	Estimate cooling loss; decide at Month 4 whether Part III remains full or descriptive.
Month 5	Validate logged-care predictions and compare allocation rankings.
Month 6	Bridge test, scale sensitivity, writing and reproducible code release.

Conclusions + Bibliography

Expected Contributions

A neighbourhood civic heat-response indicator

Daily logged-care response with uncertainty, tested against participation inequality and a static social-status benchmark.

Evidence on drought-related cooling loss

Separate canopy from low vegetation and map the distribution of total loss across Berlin.

An allocation comparison and reproducible pipeline

Calibrated logged-care predictions, transparent priority rankings, documented data integration and reusable code.

Interpretation: LST is not felt heat; logs are not all watering; response slopes are descriptive; allocation embeds a policy choice.

Selected Bibliography

Schwaab et al. (2021) Urban trees and land surface temperature in European cities. Nature Communications 12, 6763.

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Stechemesser & Wenz (2023) Inequality in behavioural heat adaptation using New York mobility data. Lancet Planetary Health 7, e798–e808.

Kirklies et al. (2024) Digital equality in co-production and platform design. Government Information Quarterly 41, 101903.

Bürkner (2017); Roberts et al. (2017) Bayesian multilevel models (brms); cross-validation for spatial and temporal data. JSS 80(1); Ecography 40.

Data: CityLAB / Gieß den Kiez · Berlin Geoportal, MSS and Umweltatlas · DWD · USGS Landsat · ESA Sentinel-2 / CCI

Thank you

Danke

Questions?

Fragen?